The Index as Hub

Price Transparency and Tacit Collusion in Ethanol Distribution

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4th International Workshop on Game Theory and Economic Applications (IWGTEA)

Transparency has two faces

- **Price transparency**: usually pro-competitive;
  - informed buyers discipline sellers;
  - regulators gain visibility.
- **But**: if rivals can costlessly observe each other's prices, the informational precondition for tacit collusion — detect & punish deviation — is satisfied *without any agreement*;
- **Evidence**: Danish ready-mix concrete (Albæk et al. 1997, +15–20% prices); Alberta electricity bid data (Brown et al. 2025).

Question

Can a *publicly disseminated price index* function as a non-firm hub sustaining tacit collusion?

Setting: the CEPEA/ESALQ ethanol index

- Since Jan. 2010: CEPEA/ESALQ (USP) + B3 publish a **daily spot price index** for hydrated ethanol, CIF Paulínia (SP);
- Constructed from executed spot transactions; released next business day;
- Reference for B3 futures settlement *and* de facto benchmark for physical supply contracts nationwide;
- Market structure: $\approx$ 300 distributors nationally, but **Vibra + Raízen + Ipiranga** $>$ 53% of hydrated ethanol sales;
- Classic Stigler (1964) conditions: few dominant players, repeated interaction, homogeneous product, public daily signal.

Why this signal is special

- The index signals *past* actions, not future commitments;
- In repeated-game language: the public monitoring signal of Green & Porter (1984) — detects deviation with a 1-day lag (3 days over weekends);
- Unlike futures prices (Allaz–Vila 1993; Mahenc–Salanié 2004), a *spot* index carries no strategic ambiguity: it just makes deviation observable.

Research question

Does the CEPEA index act as a **non-firm hub** in a hub-and-spoke structure — solving monitoring/coordination without direct communication, and without any single firm bearing legal responsibility?

What comes next

1. **Theory** — stage game, two competing mechanisms;
2. **Simulation** — agent-based model, results, robustness;
3. **Implications** — contributions, policy, future work.

The stage game

- $n = 5$ distributors at the Paulínia hub, infinite discrete time;
- Firm i chooses $p_{i,t} \in [c_i, \bar{p}]$, $c_i \sim F(\cdot)$ heterogeneous costs;
- Demand: discrete choice with partial loyalty λ;
- Profit ranking: $\pi^d > \pi^m > \pi^*$ (deviation > collusive > Nash).

The informational hub

$$\mathcal{I}_t = \frac{1}{|\mathcal{R}_t|} \sum_{i \in \mathcal{R}_t} p_{i,t}, \qquad \ell_t = \begin{cases} 1 & \text{Mon–Thu} \\ 3 & \text{Fri (weekend)} \end{cases}$$

Aggregates prices, hides identities, broadcasts to all ⇒ transforms *private* into *public* monitoring (Tirole 1988: a *facilitating practice*).

No strategic sophistication required — firms treat the index as a focal point:

$$p_{i,t+1} = (1-\alpha)\, p_{i,t} + \alpha\, \mathcal{I}_{t-\ell}$$

Proposition (Adaptive Convergence)

For any $\alpha \in (0,1)$: cross-firm dispersion $\mathrm{Var}(p_t) \to 0$ as $t \to \infty$, geometrically, **without** any punishment mechanism, beliefs about rivals, or discount factor δ.

Proof sketch: $V_{t+1} = (1-\alpha)^2 V_t$ — a contraction. Baseline $\alpha = 0.04 \Rightarrow$ half-life ≈ 8.5 periods.

Relates to adaptive learning (Fudenberg–Levine 1998) and to Calvano et al. (2020, Q-learning collusion) — but here the coordinating device is *institutional*, not algorithmic.

Mechanism II: Strategic deterrence

If the index lets firms detect & punish deviation, collusion can be a subgame-perfect equilibrium (folk theorem; Green–Porter 1984).

- **Grim-Trigger:** cooperate while $\mathcal{I}_{t-\ell} \geq p^{col} - \varepsilon$; else Nash forever;
- **Tit-for-Tat:** punish one period when the index falls below the rolling reference.

Proposition (TfT threshold)

$$\delta^{TfT}(\mathcal{I}, \ell) = R^{1/\ell}, \qquad R \equiv \frac{\pi^d - \pi^m}{\pi^m - \pi^*}$$

Index enables collusion at $\delta < 1$; longer lag ℓ requires **more** patience: $\delta^{TfT}(\mathcal{I}, 3) > \delta^{TfT}(\mathcal{I}, 1)$.

Two mechanisms, one test

- If **Mechanism I** dominates: disabling punishment (keeping the index) $\Rightarrow$ little effect on dispersion;
- If **Mechanism II** dominates: disabling punishment $\Rightarrow$ destroys coordination.

This is an empirical question. The ABM answers it.

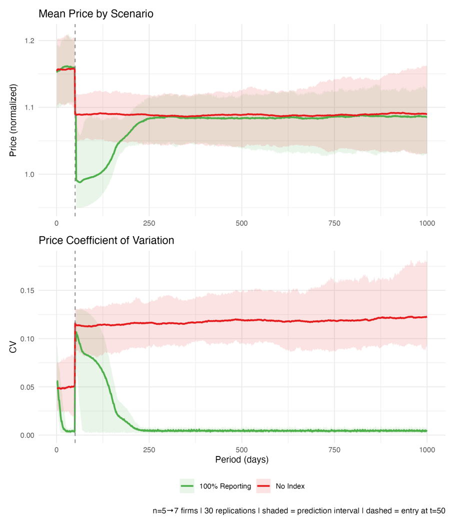
Agent-based model

- Calibration: $n = 5$ (3 majors + 2 regional) $\rightarrow$ 2 entrants at $t = 50$;
- Two scenarios: **No Index** vs. **Full Index** ($\ell = 1/3$);
- 1,000 periods, **30 replications** each;
- Firms implement *both* mechanisms: $\alpha = 0.04$ convergence + TfT ($\varepsilon = 0.05$);
- Factorial control (§4.2) selectively disables each channel.

H1–H3

H1: index $\Rightarrow$ lower CV, higher profit; **H2:** Friday lag $\Rightarrow$ more dispersion/punishment; **H3:** entry disrupts no-index more persistently.

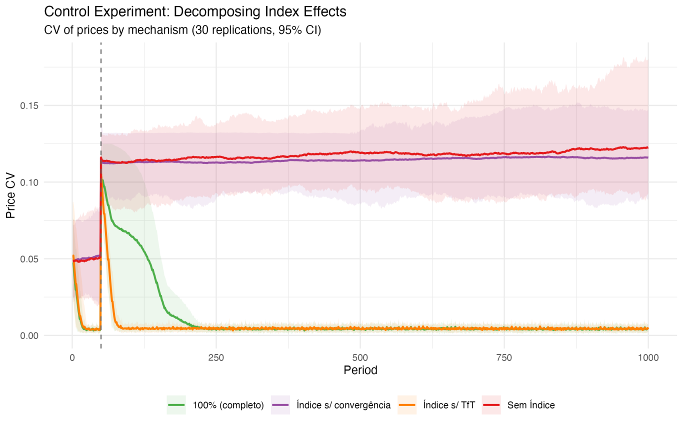
Result 1: Price coordination (H1)



Result 1: Price coordination (H1)

- $\mathrm{CV}_{100\%} = 0.005$ vs. $\mathrm{CV}_{\emptyset} = 0.119$ — **26× lower dispersion** ($p < 10^{-15}$);
- Mean price levels *similar* across regimes ($\approx$ 1.08–1.09) $\Rightarrow$ effect is coordination, not a price premium;
- Profit premium: $+4.9\%$ pooled, **not** significant ($p = 0.14$) — fragile across seeds;
- TfT **never** triggers in the stationary phase: 0 punishments / 27,000 obs.

Result 2: Which mechanism drives it?



Result 2: Decomposition – decisive

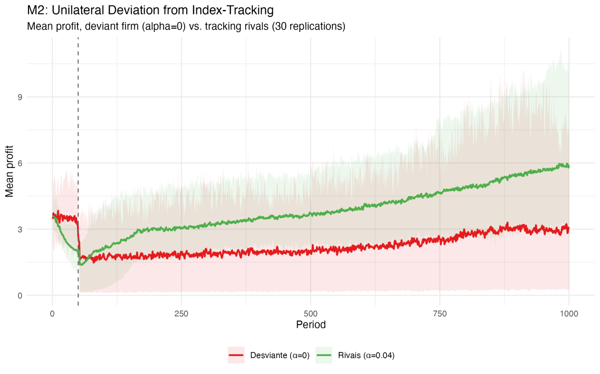
Config.	Convergence	TfT	CV
(A) No index	–	–	0.119
(B) Full model	0.04	Yes	0.004
(C) TfT only	0	Yes	0.115
(D) Convergence only	0.04	No	0.004

- **Mechanism I alone: 99.9%** of CV reduction;
- **Mechanism II alone: 3.3%** – barely different from baseline;
- Config. C profit: $\bar{\pi}_C = 1.15$ vs. $\bar{\pi}_A = 4.14$ — TfT alone **destroys ~75% of profits** via mistriggered punishment.

Central finding

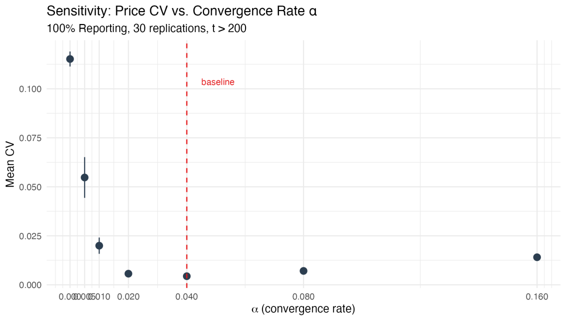
Coordination here is **behavioral**, not strategic: firms track a signal, not fear retaliation.

Is index-tracking individually rational?



- Firm forced to $\alpha = 0$ (non-tracker) vs. rivals at $\alpha = 0.04$;
- Non-tracker loses **-1.87/period** vs. tracking rivals ($p = 2.9 \times 10^{-8}$);
- Opting out is *not* profitable $\Rightarrow$ tracking is a behavioral equilibrium, not just an imposed rule.

Robustness: which α, and how confident?



- Non-monotonic: CV minimized near baseline $\alpha = 0.04$; overshoot at high α;
- **Indirect inference** vs. real CEPEA data (4,001 days, 2010–2026): data most consistent with $\alpha \approx 0.005$–0.01;
- At data-consistent α: CV reduction is **2–6**$\times$, not 26$\times$ — smaller, still substantial.

Honest caveats at the data-consistent α

- **Holds up:** CV reduction remains large & significant; Mechanism I still dominates; unilateral deviation still loses more;
- **Does NOT survive**:
  - Profit premium **reverses sign** (still n.s.);
  - Public index vs. private monitoring ranking **reverses** — at low α, private monitoring outperforms the public index.

We report this as a mixed result

Robust claim across calibrations: **price dispersion falls** under the index. Profit effects and the public-vs-private ranking are calibration-dependent.

Contributions

1. Formalizes **two competing mechanisms** for index-facilitated collusion; factorial test shows adaptive convergence dominates;
2. ABM as an **equilibrium-selection device**: theory identifies candidate mechanisms, simulation shows which one operates;
3. Direct **competition-policy** implications for CADE's fuel-market screening.

Policy implications

- Index has genuine pro-competitive value: search costs $\downarrow$, price discovery, benchmark for smaller distributors;
- But coordination emerges via **behavioral convergence alone** — no punishment to detect, no observable intent;
- Harder problem than explicit collusion for standard screens (Ragazzo–Silva filters target cartel conduct);
- Policy-relevant lever: α — publication frequency, aggregation level, signal precision;
- Contrast: Alberta's electricity regulator *restricted* public bid data in response to similar dynamics (Brown et al. 2025);
- We do not prescribe a specific intervention — we map the trade-off; the choice belongs to the regulator.

Future work

- **Natural experiment:** same distributors trade ethanol (indexed) *and* gasoline A (not indexed) — within-firm test using ANP microdata;
- Full grid search over $(\alpha, \varepsilon_{TfT}, \tau_{RL})$;
- Structural welfare model of the vertical chain.

Thank you

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